

KNOXVILLE, IA, USA

WMO: 725493

Lat: 41.299N

Lon: 93.114W

Elev: 926

StdP: 14.21

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04962

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -3.6	(c) 1.5	(d) -12.8	(e) 2.8	(f) -2.4	(g) -8.2	(h) 3.7	(i) 3.6	(j) 28.3	(k) 23.1	(l) 26.0	(m) 25.7	(n) 8.6	(o) 320	(p) 0.518

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.1	93.0	77.4	90.4	76.7	87.9	75.4	80.8	89.7	78.8	87.0	77.1	84.9	9.4	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.8	154.9	87.5	76.6	143.8	85.3	74.1	131.7	82.9	45.5	89.9	43.0	87.1	41.2	84.8	98.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 24.7	(b) 21.0	(c) 18.8	(e) -12.6	(f) 96.8	(g) 7.8	(h) 5.5	(i) -18.2	(j) 100.8	(k) -22.8	(l) 104.0	(m) -27.2	(n) 107.2	(o) -32.9	(p) 111.2		
(4) 24.7	(5) 21.0	(6) 18.8	(7) -12.9	(8) 82.7	(9) 7.7	(10) 2.2	(11) -18.5	(12) 84.3	(13) -23.0	(14) 85.6	(15) -27.3	(16) 86.9	(17) -32.9	(18) 88.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.1	24.9	27.9	40.0	52.4	63.1	72.2	76.2	74.0	67.0	53.7	41.9	30.0
	DBStd	20.48	12.39	12.95	12.51	9.44	8.32	6.29	5.92	5.59	8.35	9.61	11.91	11.33
	HDD50	2836	777	624	359	81	6	0	0	0	2	71	293	623
	HDD65	5839	1242	1040	779	389	138	14	2	4	76	368	703	1084
	CDD50	3590	0	4	48	153	411	667	811	744	512	187	49	4
	CDD65	1117	0	0	2	11	78	230	348	283	136	19	10	0
	CDH74	10276	0	0	18	139	669	2028	3462	2461	1167	156	176	0
	CDH80	3738	0	0	1	24	175	711	1409	875	388	29	126	0
Wind	WSAvg	8.8	9.8	9.6	10.4	10.5	9.2	7.8	6.4	6.0	7.3	8.9	9.7	9.5
Precipitation	PrecAvg	36.1	1.0	1.3	2.0	3.9	4.9	5.4	4.3	4.1	3.5	2.4	1.9	1.3
	PrecMax	52.9	2.2	2.4	5.4	8.0	9.6	13.0	12.4	10.0	7.2	6.1	4.9	4.7
	PrecMin	23.4	0.6	0.2	0.0	1.6	1.2	1.0	1.0	0.1	1.7	0.3	0.0	0.1
	PrecStd	7.2	0.4	0.6	1.3	1.5	2.0	2.9	2.7	2.6	1.6	1.5	1.2	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.4	65.5	76.7	82.5	90.2	93.2	99.3	94.9	92.5	83.7	102.6	63.3
		MCWB	47.3	53.3	62.1	65.7	73.5	77.3	78.4	76.5	76.6	70.3	84.0	56.8
	2%	DB	50.3	58.6	70.3	78.6	84.2	90.3	93.5	90.9	88.3	78.7	71.7	54.9
		MCWB	42.9	48.7	57.7	62.3	70.2	75.8	78.5	77.0	73.2	65.1	58.7	48.3
	5%	DB	45.3	51.6	64.1	72.9	81.2	87.6	90.5	88.1	84.3	73.3	63.3	49.7
		MCWB	39.7	44.6	54.9	59.3	67.2	74.8	77.9	76.3	72.3	62.6	54.0	45.0
	10%	DB	40.6	45.4	59.0	68.4	77.3	84.0	87.8	84.3	81.2	69.8	57.3	45.1
		MCWB	36.1	40.1	50.2	56.5	65.6	73.1	76.5	74.2	69.9	60.5	49.4	40.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.2	57.0	64.4	69.0	76.5	80.6	84.6	82.5	78.6	73.3	81.2	58.1
		MCDB	54.7	63.1	73.0	79.9	85.7	89.4	95.0	91.3	86.8	81.8	99.5	62.8
	2%	WB	43.5	49.3	60.5	64.7	72.7	78.3	81.9	79.9	76.2	68.7	60.0	50.7
		MCDB	49.9	57.3	67.1	74.1	82.0	87.5	90.8	88.0	84.8	75.2	68.1	53.3
	5%	WB	39.9	44.6	56.6	61.2	69.8	76.3	79.5	77.8	73.9	65.4	55.2	44.3
		MCDB	45.5	50.8	63.9	70.7	78.2	84.6	87.7	84.9	81.7	71.5	62.5	48.3
	10%	WB	36.3	40.3	50.5	58.2	67.3	74.4	77.6	75.5	71.8	61.3	50.4	40.2
		MCDB	40.3	45.6	58.2	66.7	75.1	82.2	85.2	83.0	79.1	67.9	57.0	44.9
Mean Daily Temperature Range	5% DB	MDBR	15.7	16.2	18.0	20.2	18.8	17.8	18.1	17.9	20.3	18.9	17.6	14.6
		MCDBR	21.8	23.9	26.0	27.3	22.5	20.2	20.4	20.3	22.2	23.1	26.0	19.7
		MCWBR	16.2	16.5	16.0	15.9	11.9	10.9	10.3	10.9	11.0	13.2	17.1	14.9
	5% WB	MCDBR	21.3	21.2	22.4	23.7	19.8	18.1	18.3	17.5	19.0	19.3	24.1	17.8
		MCWBR	16.1	15.8	15.1	16.1	12.2	11.6	11.4	11.3	10.4	14.1	17.5	14.8
Clear-Sky Solar Irradiance	taub	0.287	0.314	0.333	0.386	0.409	0.420	0.436	0.419	0.383	0.341	0.300	0.290	
	taud	2.428	2.353	2.394	2.262	2.266	2.281	2.225	2.292	2.342	2.459	2.504	2.439	
	Ebn at Noon	277	284	291	280	275	272	266	267	270	271	270	265	
	Edh at Noon	25	31	34	42	42	42	44	40	35	28	23	23	
All-Sky Solar Radiation	RadAvg	610	924	1200	1508	1774	1969	1985	1727	1466	983	668	514	
	RadStd	59	66	108	111	137	194	147	106	107	112	57	49	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page